

Week 11

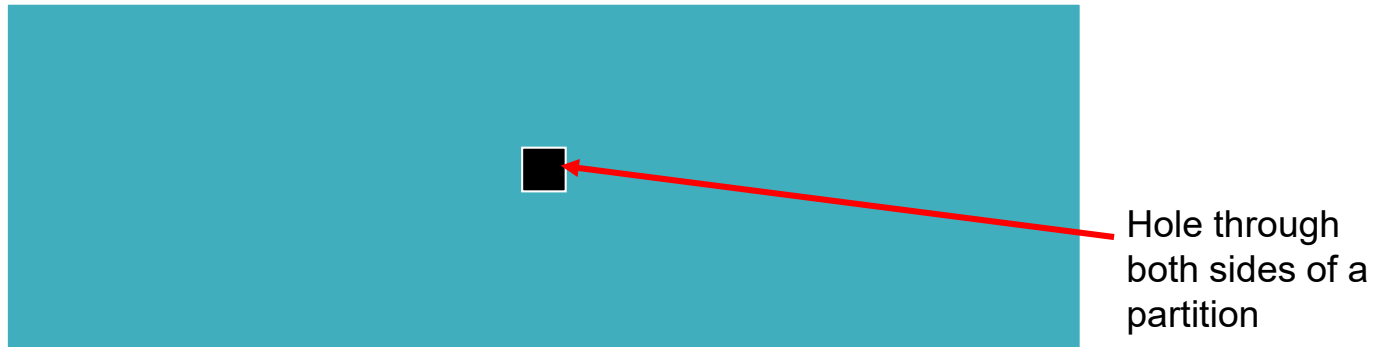
Noise Barriers

Spring 2021
Kurt Neiswender, AIA
College of Architecture and Design
Lawrence Technological University

EFFECTS OF NOISE LEAKS

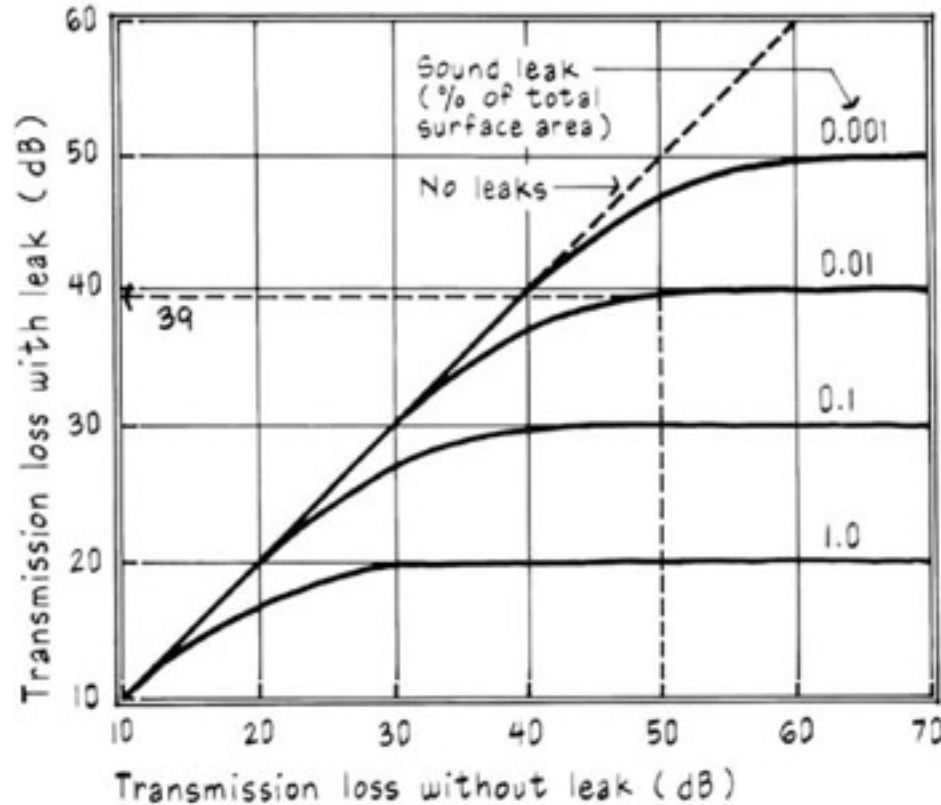
OPENING IN BARRIER:

Example



- Barriers can be highly effective for blocking noise. Their performance is often limited by gaps and openings that allow noise to leak through (flanking paths).
- Even if the area of this leak is very small compared to the entire panel, there is a substantial reduction in TL.

EFFECTS OF NOISE LEAKS



- For any partition, the transmission loss with a leak can be estimated by knowing the percentage of the leak to the total surface area of the partition.
- For example, a partition with a TL of 50 dB without leaks would have a TL of 39 dB if it has a leak of only 0.01% - a reduction of 11 dB in the TL!

AIRBORNE SOUND ISOLATION (*BLOCKING SOUND*)

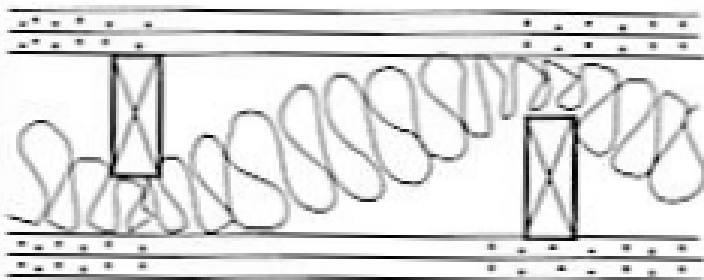
BUILDING
CONSTRUCTION
CONSIDERATIONS

LIGHTWEIGHT GYPSUM BOARD ASSEMBLIES

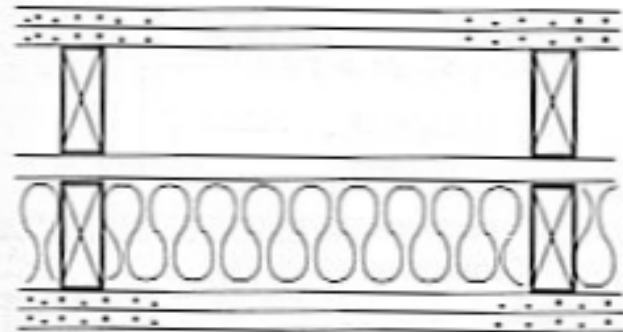
- Lightweight gypsum board assemblies consist of gypsum boards on either side of either wood or metal studs.
- The partition can have single or multi-layered board on each side
- The cavity can be empty or contain a porous absorber

LIGHTWEIGHT GYPSUM BOARD ASSEMBLIES

- STC values vary between 37 and 57 for gypsum board assemblies *without cavity absorption* and 40 and 63 for gypsum board assemblies *with cavity absorption*.
- Staggered studs are a slight improvement to resilient channel
- Double studs give the best STC values, but they must be independent (no bracing or ties between stud rows).

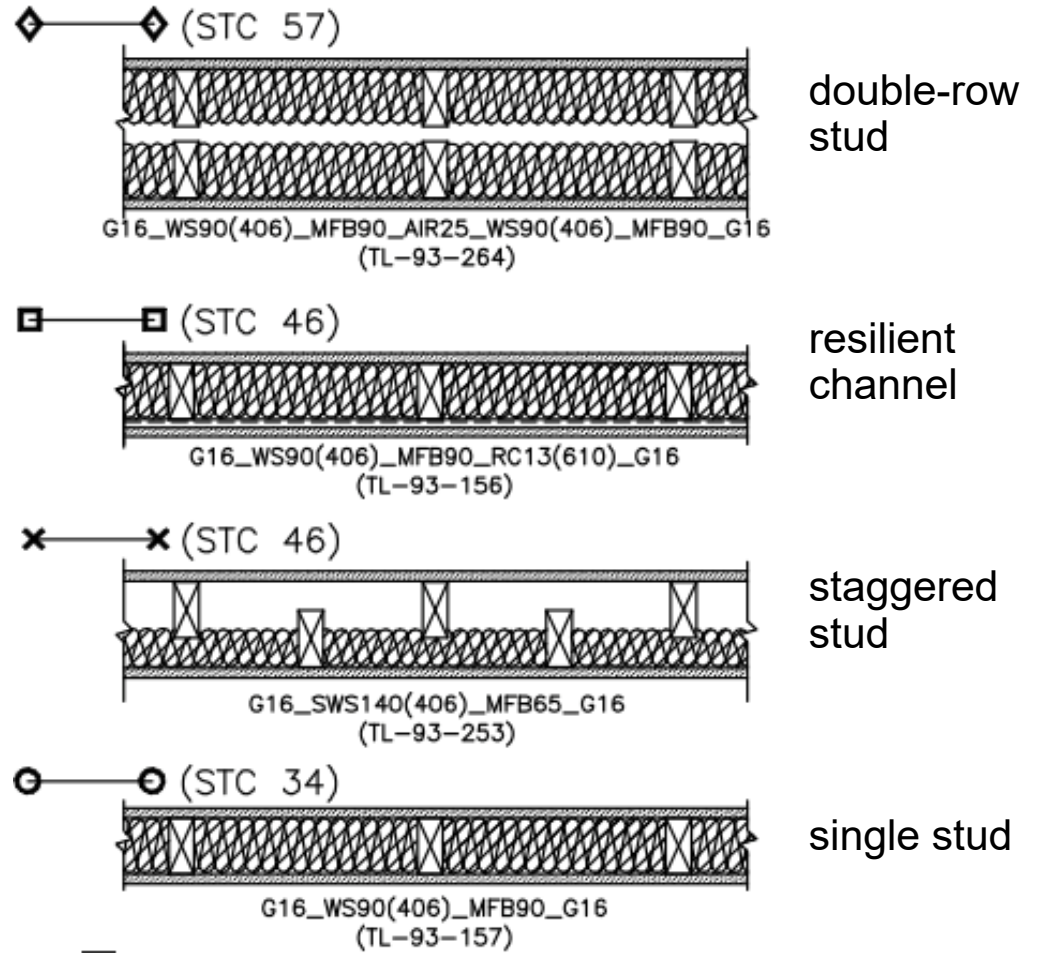
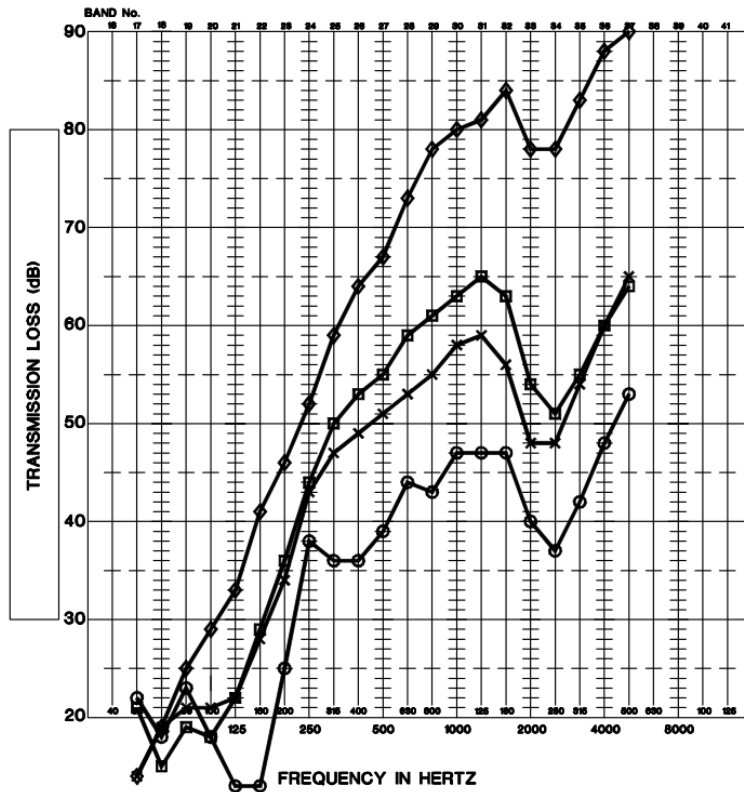


Staggered stud wall
Approx. STC 55



Double stud wall assembly
Approx. STC 63

LIGHTWEIGHT GYPSUM BOARD ASSEMBLIES



LIGHTWEIGHT GYPSUM BOARD ASSEMBLIES

- Resiliently mounted gypsum board – decouples one leaf from studs
 - Resilient channel (RC)
 - Difficult to install properly – short circuited by fasteners between GWB and studs
 - Not what it used to be (except for Dietrich RCSD RC Deluxe) – other test data is for a discontinued product
 - Resilient clips
 - Neoprene used for isolation
 - Relatively new product
 - Thicker than resilient channel
 - Not as easy to short circuit

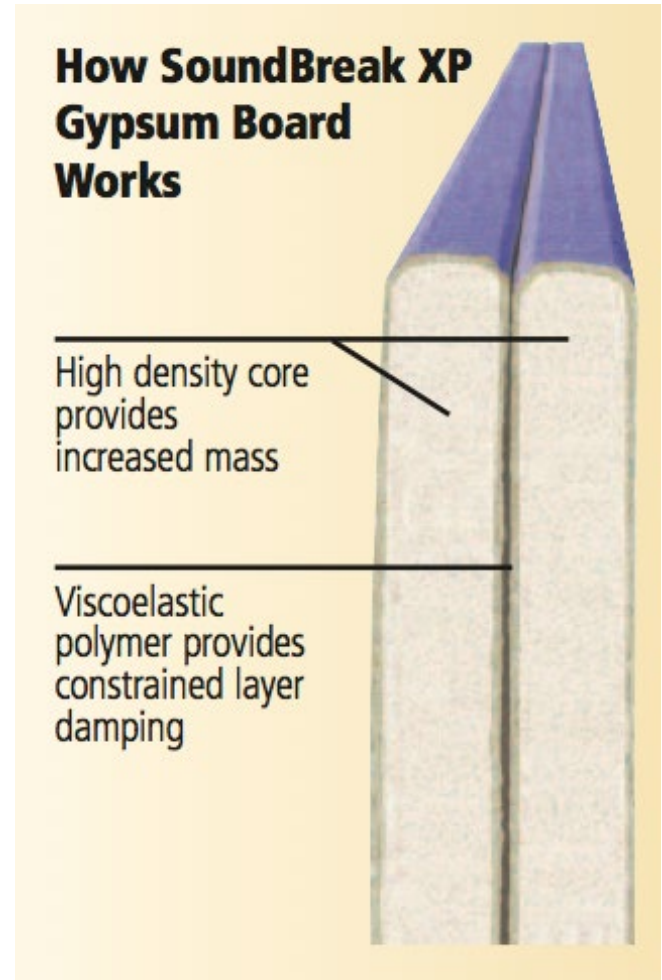


*RC Deluxe™
RCSD-STC Rating up to 61*



LIGHTWEIGHT GYPSUM BOARD ASSEMBLIES

- Internally damped gypsum board or plywood
 - *Damping layer sandwiched in gypsum board or plywood acts like a shock absorber in your car → vibrations are quickly dissipated*
- Large increases in STC are generally due to almost eliminating the coincidence dip via damping and an increase in low frequency performance from additional mass
- Best performance when directly attached to studs



LIGHTWEIGHT GYPSUM BOARD ASSEMBLIES

- *STC values increase:*

- Slightly as stud spacing increases (24" o.c. vs. 16" o.c.) because there is less stiffness and fewer contact points in the assembly
- Slightly with larger studs (creating a deeper air cavity and more decoupling of gypsum boards) -particularly at low frequencies
- Significantly for staggered studs and double studs (decoupling of gypsum board surfaces)
- Significantly with resilient mounting of gypsum board (decoupling, which reduces vibration transmission)
- Significantly for damped materials (reduces bending natural frequency effects and can add mass)
- Significantly with lighter gauge (higher number) steel studs (decreasing stiffness)

- The same strategies for increasing STC for walls also work for floor/ceiling assemblies

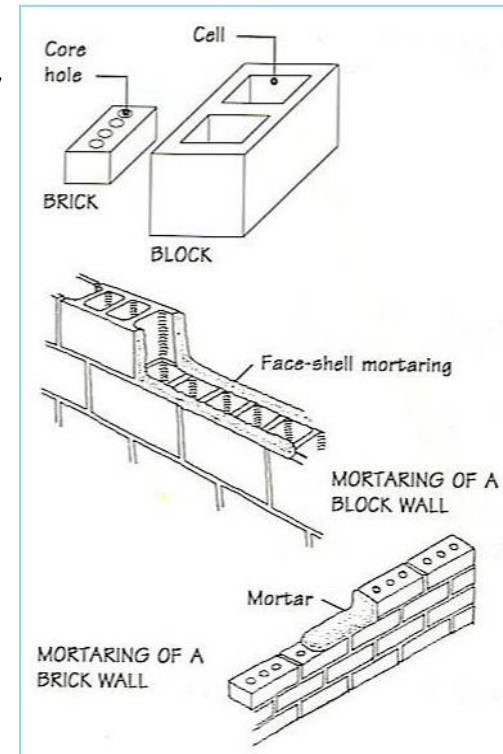
CONCRETE WALLS AND SLABS

- Because of **high stiffness** and **thickness** of poured concrete walls, *the coincidence dip takes place at low frequencies*, sometimes below the STC range
 - 4" panel - coincidence dip around 200 Hz (still within STC range)
 - 8" panel - coincidence dip around 100 Hz (out of STC range)
- In STC calculation range, the TL increases approximately 5 dB per one-third octave band



MASONRY WALLS

- Masonry walls are composed of blocks or bricks and mortar – *typically single-leaf partitions*
- Single-Leaf Masonry Walls TL improvement options
 - Use **thicker & heavy weight** block (weight can vary by mfg)
 - **Seal both sides** with 2-3 coats of latex block filler, thick epoxy paint, plaster, or gypsum board on furring (if resiliently mounted, air gap should be 2" or more and filled with fiberglass)
 - Make sure **all joints are tightly filled**
 - **Fill cells** with sand or mortar/grout (sand settles though)
 - Surface mount electrical services
 - Use independently supported double masonry wall for very high TL



SOUND INSULATING DOORS

- **Ordinary (unsealed) doors** have gaps typically about 1/4" wide at all edges. If not sealed, STC values are 22 or less.
- **Solid core doors** have somewhat higher STC ratings than hollow core doors
- Sound insulating doors loosely fall into two categories: fully sealed doors (mid range STC 25-42) and manufactured acoustical doors (higher STC 41-64) or acoustical doors in tandem (highest STC 57+)

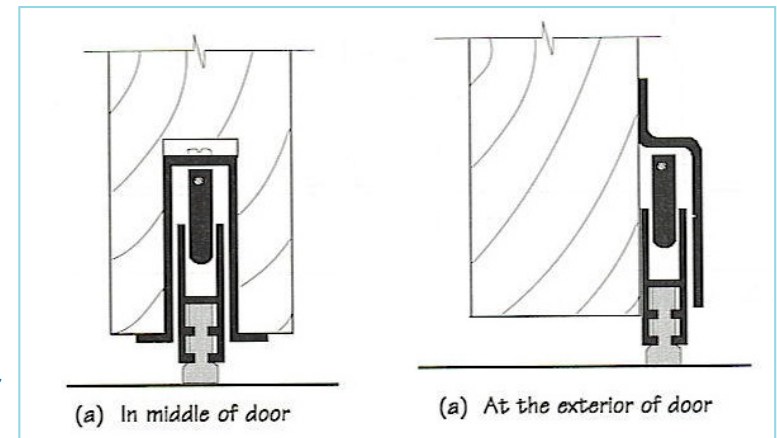
Door type	Surface mass kg/m ² (lb/ft ²)	STC (Unsealed)	STC (Sealed)
Hollow core wood door	7.0 (1.5)	17	20
Solid core wood door	20.0 (4.1)	20	28
Hollow core steel	25 (5.1)	20	30
Two solid core wood or two hollow core steel doors with 70 mm (2.8 in.) space	20.0 (4.1) each	28	40
Two solid core wood or two hollow core steel doors with 230 mm (9 in.) space	20.0 (4.1) each	42	50

SOUND INSULATING DOORS

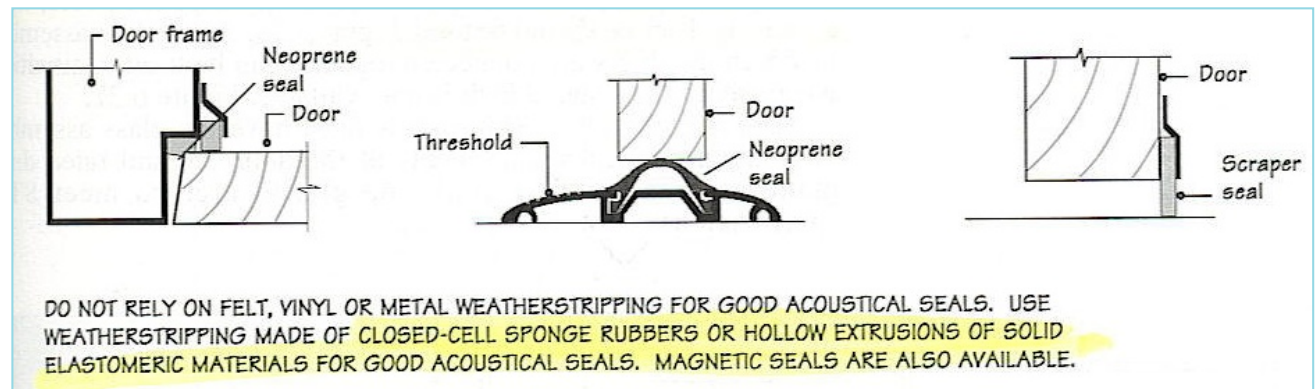
Sealed Doors

- The *entire perimeter* of a door must be sealed to improve the TL/STC of the door. Neoprene is commonly used. Seals may wear over time.

Automatic Door
Bottom Seals



Scraper seal is least preferred of shown options for sealing a door bottom.



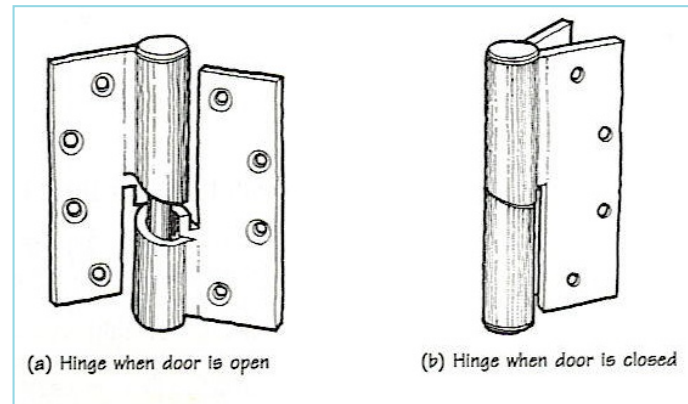
SOUND INSULATING DOORS

Acoustical Doors

- Special construction by manufacturers range in STC from 41 to 64. **Seals, hinges and leafs** are all **specially designed for high STC values**

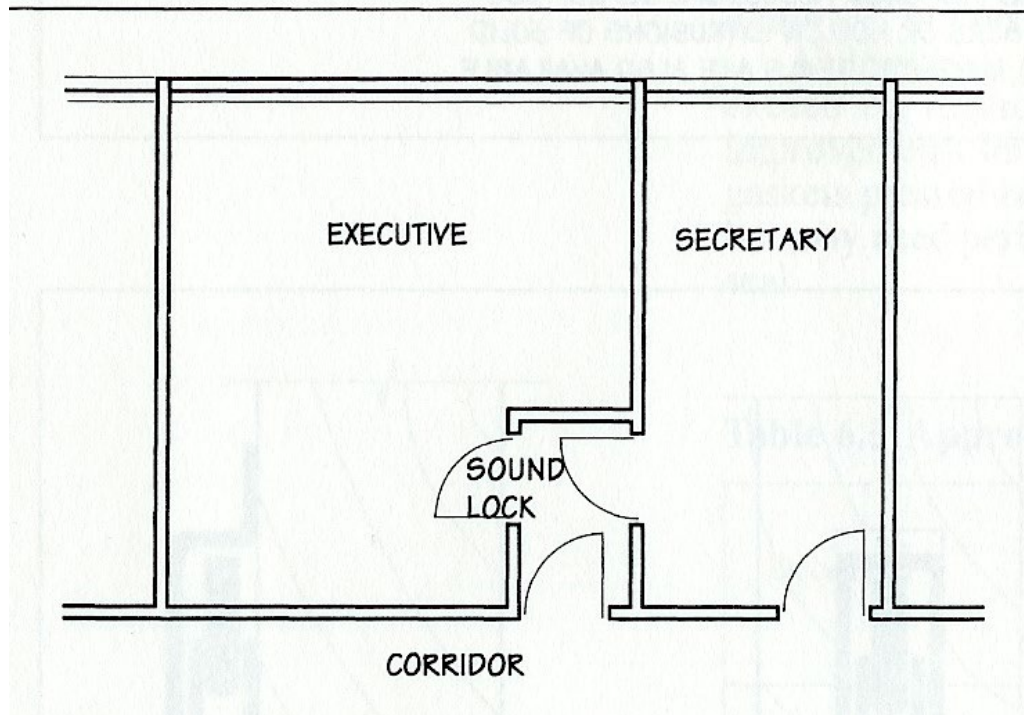


This acoustic door is mounted with special (“cam lift”) hinges which move the door downward as it closes. Gaskets around the door help to seal it from sound leaks and flanking. Combination foam and magnetic gaskets often used.



SOUND INSULATING DOORS

- Double doors, in the same frame (provide other means of egress) or through creating a vestibule (sound lock), are also used to increase STC values.



SOUND INSULATING WINDOWS

- Standard single and double glazing have **relatively low STC values**
 - Single panes standard glass STC 29-36
 - Single panes laminated glass STC 35-41
 - Double glazing with small air gap STC 28-36
 - Double glazing with <1" air gap STC 38-47
 - Double laminated glazing with small air gap STC 39-42
 - Double laminated glazing with <1" air gap STC 45-51+
- *Small air gaps (less than 1 ") in double glazed unit can result in a lower STC than a single glazed unit with thicker laminated glass.*
- Operable window have lower STC ratings (3 to 5 points below non-operable) in STC rating.
- Use neoprene gaskets around glass perimeter

SOUND INSULATING WINDOWS

- **Manufactured acoustical windows** have STC ratings of 37-59+. They tend to have a 4" to 8" air gap, have fixed panes, have high quality neoprene seals, use different thicknesses of glass, use laminated glass and have a sound absorbing material around the interior perimeter.
 - Install strictly per manufacturer's instructions to not degrade STC.



OVERALL TRANSMISSION LOSS OF AN ASSEMBLY

Assemblies or composite constructions can be thought of as the grouping of several partition elements having different TL values.

e.g., a gypsum board wall containing a door and two windows

How do you determine the overall TL for an assembly?

One of the following methods:

1. Equation Method
2. Graphic Method: Egan, p. 191 “TL Chart for Composite Constructions”

OVERALL TRANSMISSION LOSS OF AN ASSEMBLY

Equation Method (Egan p. 188-190):

$$TL_{composite} = 10 * \log \left(\frac{\sum S}{\sum \tau * S} \right)$$
$$= 10 * \log \left(\frac{\sum S}{\sum \tau_1 * S_1 + \tau_2 * S_2 + \dots + \tau_n * S_n} \right)$$

To calculate the transmission coefficient, τ :

$$TL = 10 \log (1/ \tau)$$

$$\log^{-1}(TL/10) = (1/ \tau)$$

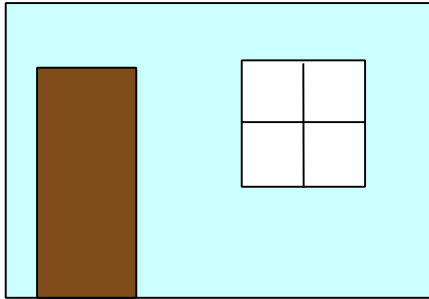
$$10^{(TL/10)} = (1/ \tau)$$

$$10^{(-TL/10)} = \tau$$

OVERALL TRANSMISSION LOSS OF AN ASSEMBLY

Practice (Equation Method):

$$TL_{\text{composite}} = 10 \log(\Sigma S / \Sigma \tau S)$$



$$\begin{aligned} S_{\text{door}} &= 21 \text{ sq. ft.} \\ S_{\text{window}} &= 16 \text{ sq. ft.} \\ S_{\text{gyp}} &= 75 \text{ sq. ft.} \end{aligned}$$

$$\begin{aligned} TL_{\text{door}} &= 25 \text{ dB} \\ TL_{\text{window}} &= 36 \text{ dB} \\ TL_{\text{gyp}} &= 49 \text{ dB} \\ &(\text{at } 500 \text{ Hz}) \end{aligned}$$

1. Calculate all transmission coefficients:

$$\begin{aligned} \tau_{\text{door}} &= 10^{(-TL/10)} \\ \tau_{\text{door}} &= 10^{(-25/10)} \\ \tau_{\text{door}} &= 0.0032 \end{aligned}$$

$$\begin{aligned} \tau_{\text{window}} &= 10^{(-TL/10)} \\ \tau_{\text{window}} &= 10^{(-36/10)} \\ \tau_{\text{window}} &= 0.00025 \end{aligned}$$

$$\begin{aligned} \tau_{\text{gyp}} &= 10^{(-TL/10)} \\ \tau_{\text{gyp}} &= 10^{(-49/10)} \\ \tau_{\text{gyp}} &= 0.000013 \end{aligned}$$

OVERALL TRANSMISSION LOSS OF AN ASSEMBLY

$$TL_{\text{composite}} = 10 \log(\Sigma S / \Sigma \tau S)$$

$$\Sigma S = (21 + 16 + 75) \text{ sq. ft.} = 112 \text{ sq. ft.}$$

$$\Sigma \tau S = \tau_{\text{door}} S_{\text{door}} + \tau_{\text{window}} S_{\text{window}} + \tau_{\text{gyp}} S_{\text{gyp}}$$

$$\begin{aligned} \Sigma \tau S &= (0.0032 * 21 + 0.00025 * 16 + .000013 * 75) \text{ sq. ft.} \\ &= 0.072 \text{ sq. ft.} \end{aligned}$$

$$TL_{\text{composite}} = 10 \log (112 \text{ sq. ft.} / 0.072 \text{ sq. ft.})$$

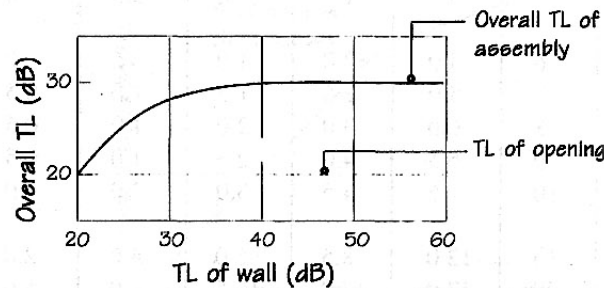
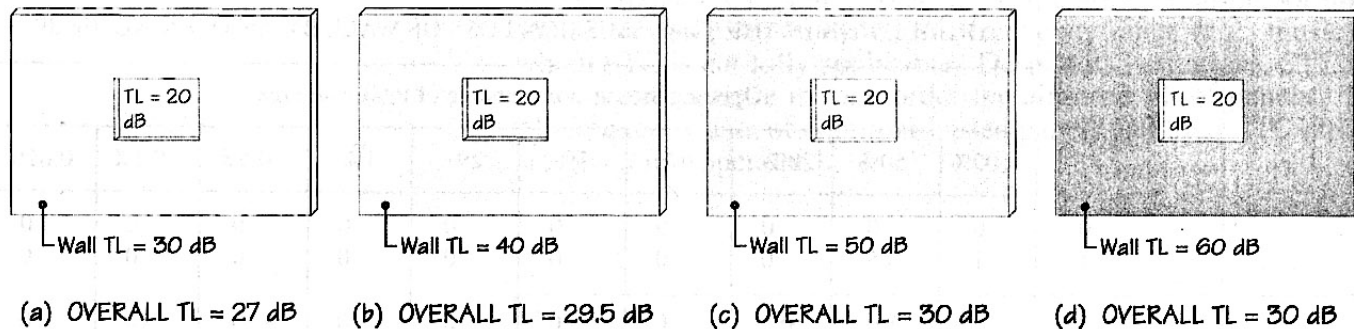
$$TL_{\text{composite}} = 10 \log (1556)$$

$$TL_{\text{composite}} = 32 \text{ dB at 500 Hz}$$

OVERALL TRANSMISSION LOSS OF AN ASSEMBLY

- For a constant size element in a partition with increasing TL, what is the effect on the composite TL?

What can we conclude from this comparison?



(e)

OVERALL TRANSMISSION LOSS OF AN ASSEMBLY

Conclusions:

- A partition with an *acoustically weak element will have a TL closer to the TL value of the weak element*, particularly noticeable when there is a large discrepancy between the TL values
- If a weak element exists in a wall, there is an upper limit to the TL that is possible. Increasing the TL of the strong element has diminishing returns.

SOUND LEAKS & FLANKING TRANSMISSION

- Air leaks are sound leaks
- Small leaks can significantly reduce the partition STC
- Sound leaks result from air leakage
- To prevent:
 - Stagger gypsum board layers when using multiple layers
 - Caulk drywall interfaces (top & bottom) with acoustic sealant (non-hardening, non-drying, dense) and backer rod if gaps are large
 - Seal deck flutes at top of partition
 - Seal the rear of outlet and data boxes
 - Back medicine cabinets, towel dispensers or any recessed object with a sealed gypsum board enclosure.
 - Locate outlet and data boxes on either side of a wall at least one stud cavity away from each other.
 - Doors and windows must be gasketed
 - Masonry joints must be completely filled and CMU painted
 - Seal all mechanical and electrical partition penetrations airtight

SOUND LEAKS & FLANKING TRANSMISSION

The TL for a leak is 0 dB. For a composite TL calculation, treat the area of the leak as if it were another element in the partition, such as a window or door

Practice: A composite STC of 44 was calculated for a CMU partition with a solid core steel door with no gaps (S=250 sf). Add a ½" x 36" gap at the bottom of the door. What is the new composite STC rating?

$$\text{Wall } \tau = 10^{(-0.1\text{TL})} = 10^{(-0.1 \cdot 44)} = 3.98 \times 10^{-6}$$

$$\text{Gap } \tau = 10^{(-0.1\text{TL})} = 10^{(-0.1 \cdot 0)} = 1$$

$$\text{TL}_{\text{composite}} = 10 \log(\Sigma S / \Sigma \tau S)$$

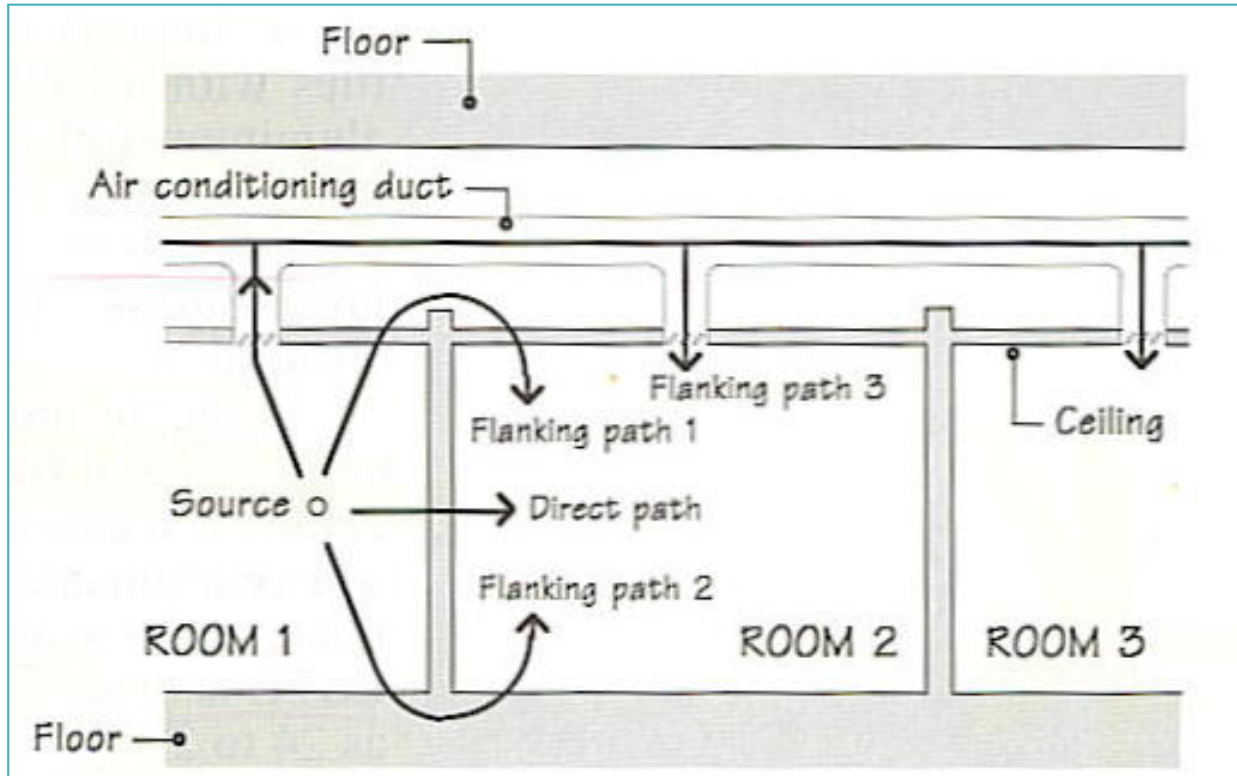
$$\text{TL}_{\text{composite}} = 10 \log [250.125 / (1 \cdot 0.125 + 3.98 \times 10^{-6} \cdot 250)]$$

$$\text{TL}_{\text{composite}} = 33 \text{ dB}$$

That's a degradation of 11 dB!

SOUND LEAKS & FLANKING TRANSMISSION

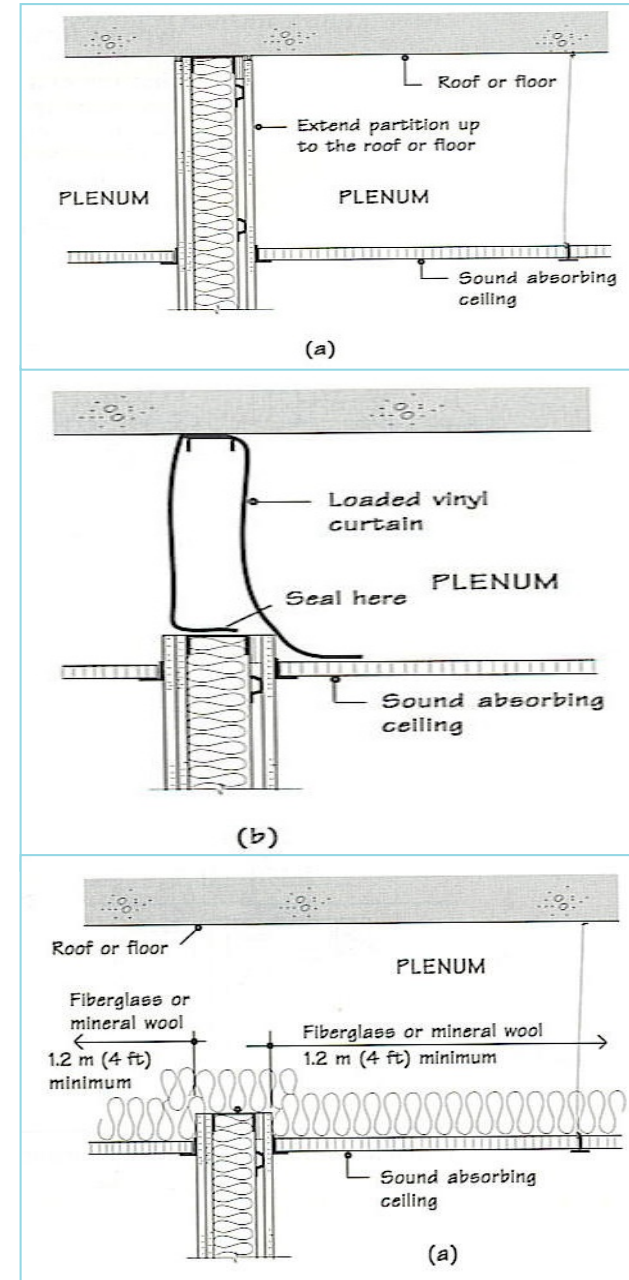
- There can be additional sound travel through other pathways, e.g., through ductwork, over partitions, through floor (structure-borne). This is called **flanking transmission**



SOUND LEAKS & FLANKING TRANSMISSION

To reduce transmission through these flanking paths:

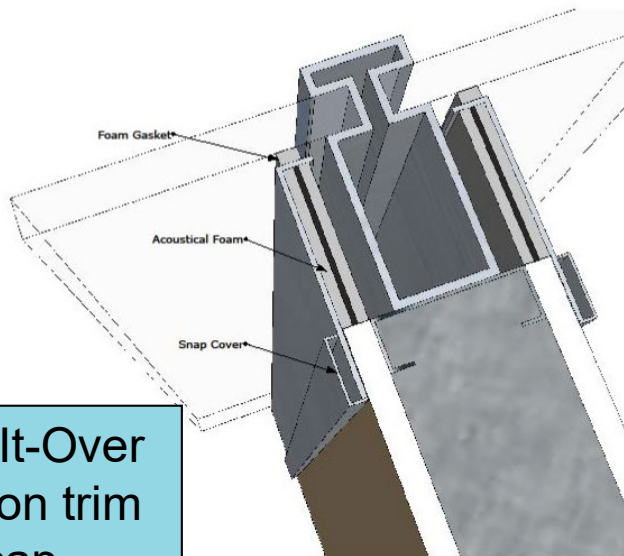
- **Extend walls to deck and seal airtight** (best option for highest TL). STC 48+
- **Add a loaded vinyl barrier** over the short wall (use in conjunction with fiberglass batt above ceiling). Suggested for retrofit. STC 35-40
- **Structural break** in the floor at the partition
- **Acoustically treat the ductwork** (we will examine this in a later lecture)
- **Replace ceiling tiles** with a high CAC tile STC 40
- Add a min. of 6-inch thick, 3 lb. **density fiberglass over tiles**. STC 35
- **Keep openings** in the ceiling (lights, sprinklers, plenum return openings, etc.) **away from perimeter walls** or acoustically treat



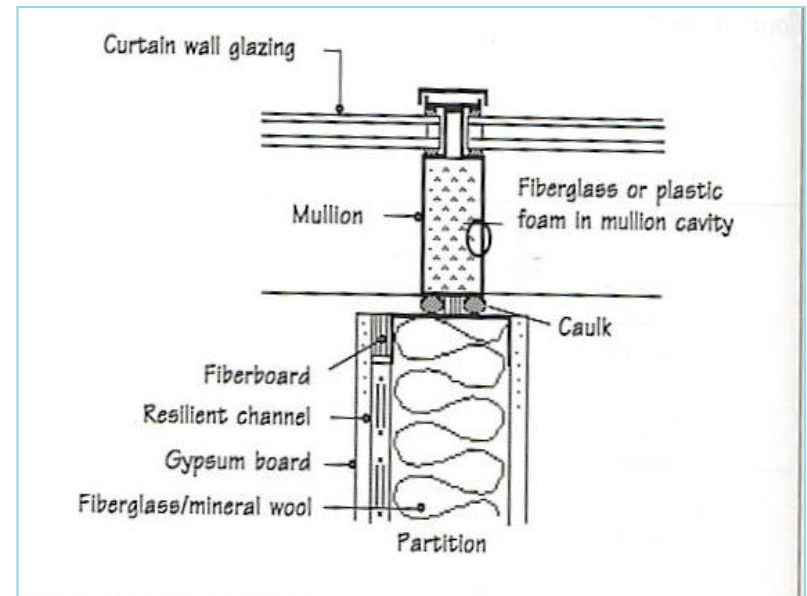
SOUND LEAKS & FLANKING TRANSMISSION

■ Mullion/wall intersections – typically very weak joints acoustically

- Design partition locations such that they meet mullions – jog the wall if you have to
- Stuff gap with fiberglass or foam and caulk airtight. A neoprene seal can be substituted.
- Expandable sheet metal cseal airtight on sides, top, bottom and at slip joint
- osures
- Mullion trim cap



Mull-It-Over
mullion trim
cap



Structure-borne Sound Insulation

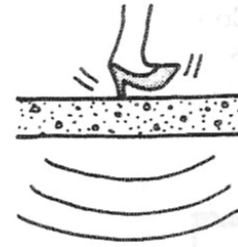
Impact noise

Structure-borne Sound Insulation

- Structure-borne sound results from two types of vibration input
 - Impact – describes a force that occurs for a short duration
 - Examples: walking, jogging, dancing, basketball, bowling, wheeled equipment & furniture, door slam
 - Vibration – produced by machinery and equipment, usually with rotating parts
 - Examples: HVAC, fans, pumps, generators, etc.

Impact Sound Insulation

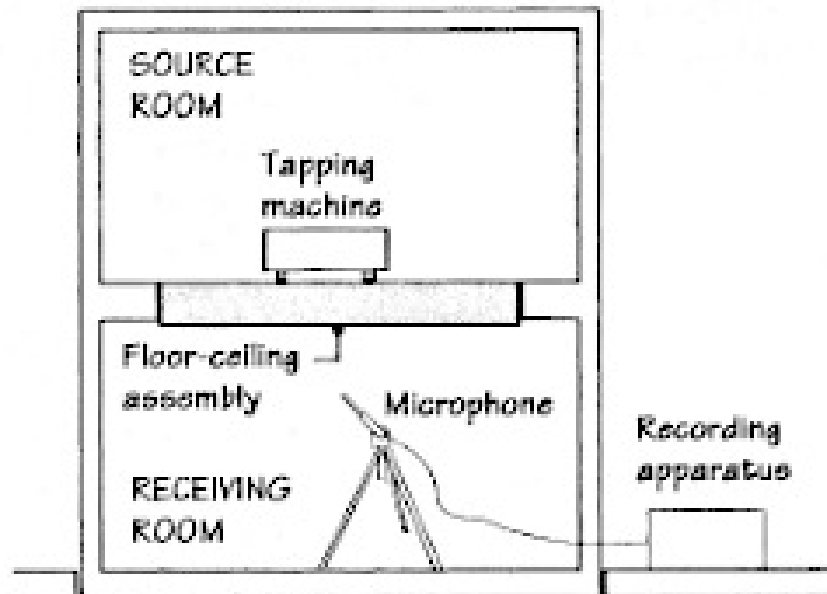
- Primarily required of floors
 - Door slams are a notable exception
 - Roof sometimes an issue for rain noise
- To insulate for impact sound, construction primarily has to be
 - **Heavy** – massive building elements are more difficult to vibrate
 - **Resilient** – energy absorption from compliant elements



Impact Insulation Class (IIC)

Impact Insulation Class testing

- Floor test assembly is installed between two test rooms
- Calibrated tapping machine with hammers is used for impact source (but hammers are not like people walking or rolling carts)



Impact Insulation Class (IIC)

Impact Insulation Class testing

- Measured in 1/3-octave bands from 100 – 3150 Hz
- *Similar to STC but inverted*
- Rating number has no units (ex. IIC 34, not IIC 34 dB)
- *Higher IIC values mean a better sound insulating floor*
- IIC is not a satisfactory indicator of “boominess” on light weight (wood construction) floors

Impact Insulation Class (IIC)

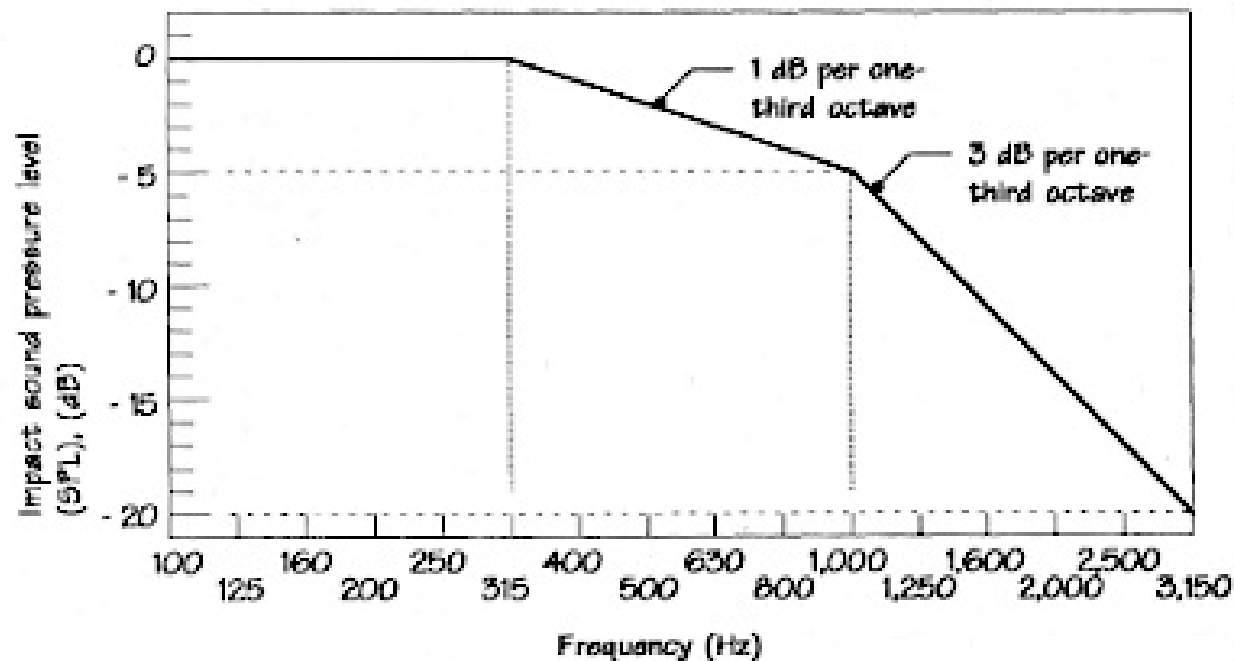
IIC Standards

- **Laboratory tested IIC:** ASTM E492 - *Standard Test Method for Laboratory Measurement of Impact Sound Transmission Through Floor-Ceiling Assemblies Using the Tapping Machine*
 - “The tapping machine itself, however, is not designed to simulate any one type of impact, such as produced by male or female footsteps.”
 - “Because of its portable design, the tapping machine does not simulate the weight of a human walker. Therefore, the creak or boom of a limber floor assembly caused by such footstep excitation may not be reflected in the single number impact rating derived from test results obtained by this test method.”
- ASTM E989 – *Classification for Determination of Impact Insulation Classes (IIC)*
 - “It does not deal with low frequency sounds below 100 Hz that are typically generated below lightweight joist floors when they are walked on.”
- **Field tested IIC (FIIC):** ASTM E1007 - *Standard Test Method for Field Measurement of Tapping Machine Impact Sound Transmission Through Floor-Ceiling Assemblies and Associated Support Structures*

Quotes from respective standards

Impact Insulation Class (IIC)

Impact Insulation Class contour



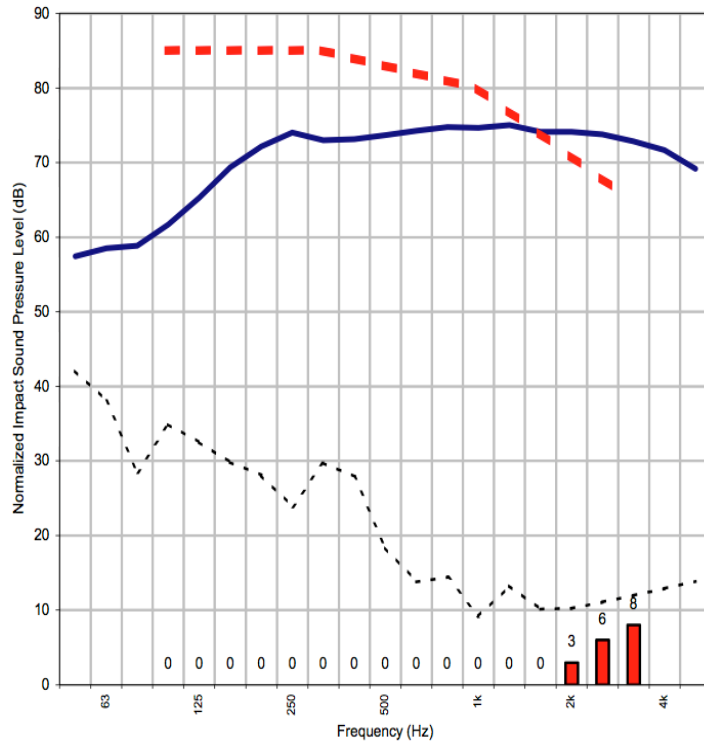
Impact Insulation Class (IIC)

Impact Insulation Class calculation procedure

- Criteria that must be met when positioning the IIC contour over the measured data
 - The sum of the deficiencies cannot exceed 32 dB
 - The maximum deficiency at any single 1/3 octave band cannot exceed 8 dB
 - Deficiencies lie above the curve!
- Read the dB level where the contour crosses 500 Hz
- Subtract this level from 110 dB to arrive at IIC rating

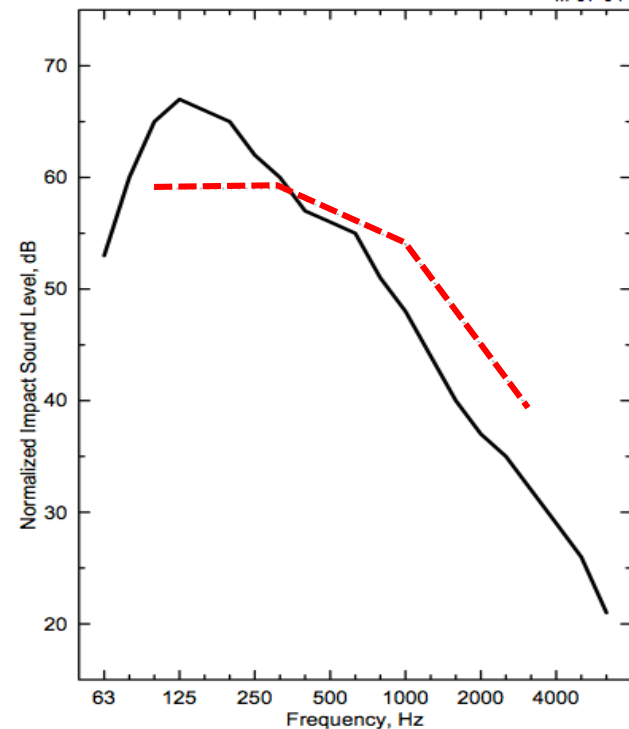
Impact Insulation Class (IIC)

Impact Insulation Class contour



IIC 27

6" concrete slab

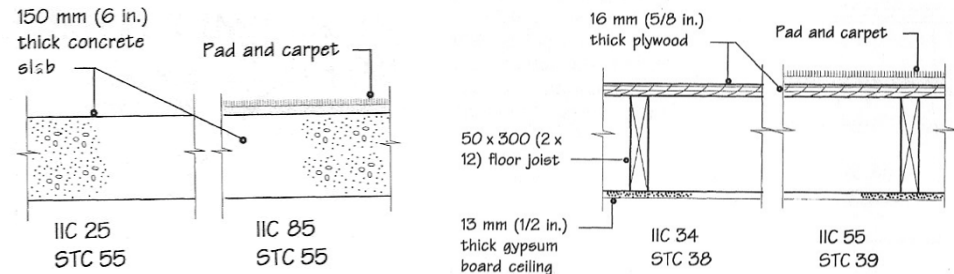


IIC 53

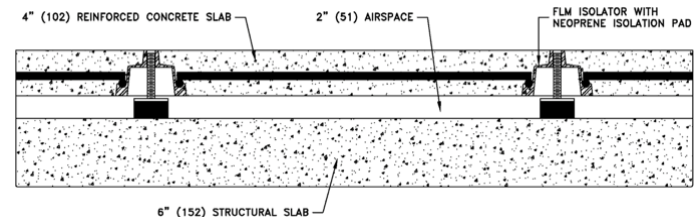
6" concrete slab topped with tile & vibration isolation underlayment

Strategies for Controlling Impact Insulation

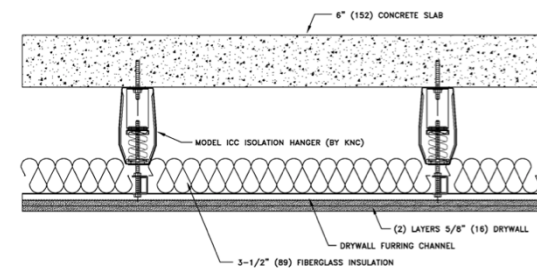
- Soft or resilient floor covering (image source: Mehta)



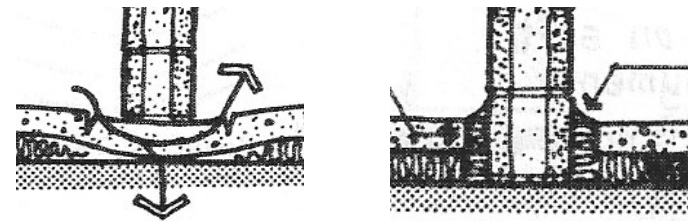
- Resiliently supported floor (commonly called a floating floor) (image source: Kinetics Noise Control website)



- Resiliently supported ceiling (commonly called a barrier ceiling) (image source: Kinetics Noise Control website)



- Structural discontinuity in floor and ceiling (reduces flanking transmission through the structure) (Egan)



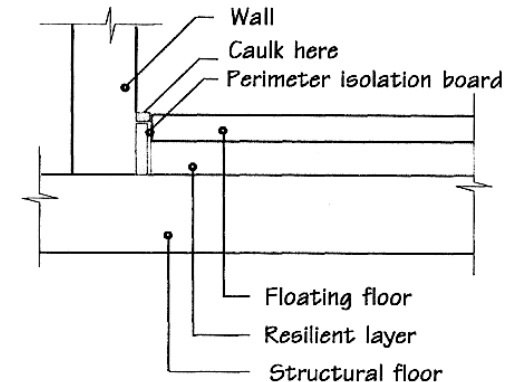
Soft or Resilient Floor Covering

- Weaken impact at the source
 - Carpet with a pad is an excellent method
 - ex. 6" concrete slab has an IIC 25 rating, 6" concrete slab with carpet and pad has an IIC 85 rating (A 60 point difference is astronomical!)
 - ex. For the more flexible floor case, a wood floor/drywall ceiling construction can be improved from IIC 34 to IIC 55 with carpet and pad
 - *The most effective & least expensive method of reducing impact sounds*
 - Cork, rubber and vinyl flooring also provide some impact insulation improvement
 - Have virtually NO effect on increasing STC

Floor	Improvement in IIC
Pad and carpet on:	
Wood floor	20
Concrete floor	60
Vinyl, rubber, etc., on:	
Wood floor	5
Concrete floor	7

Floating Floors

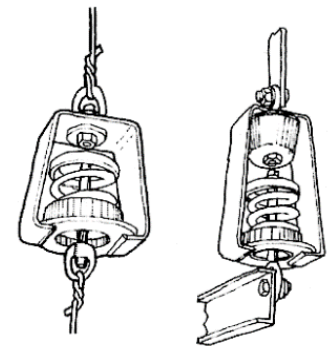
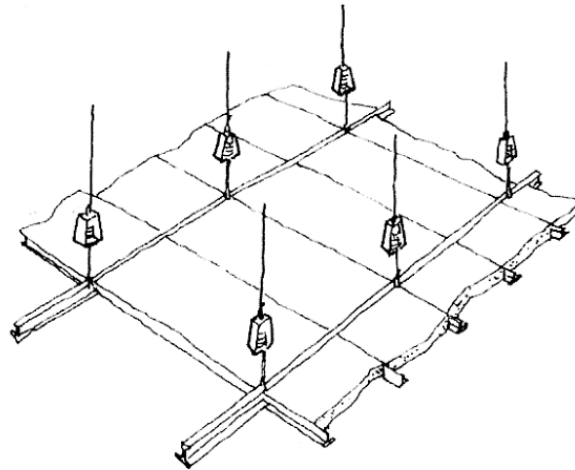
- If a hard surfaced floor exists and/or high IIC is required, a *floating floor* is used for impact insulation
 - Bonus: STC can be increased in addition to IIC
 - Resilient materials include neoprene, compressed fiberglass, honeycomb resilient floor board, springs
 - A thicker isolator is better



Floating floor	IIC	STC
Wood floating floor on:		
Wood structural floor	52	58
Concrete structural floor	64	62
Concrete floating floor on:		
Wood structural floor	58	60
Concrete structural floor	74	62

Resiliently Attached Ceilings (Barrier Ceilings)

- **Barrier ceiling system:** one or more layers of drywall on framing supported by resilient mounts (spring or neoprene)
- Must have all penetrations of barrier sealed airtight
 - Can install another grid below for mechanical and electrical attachments
- Beneficial to put fiberglass in the air space between barrier and decking above.
- Increases both IIC and STC

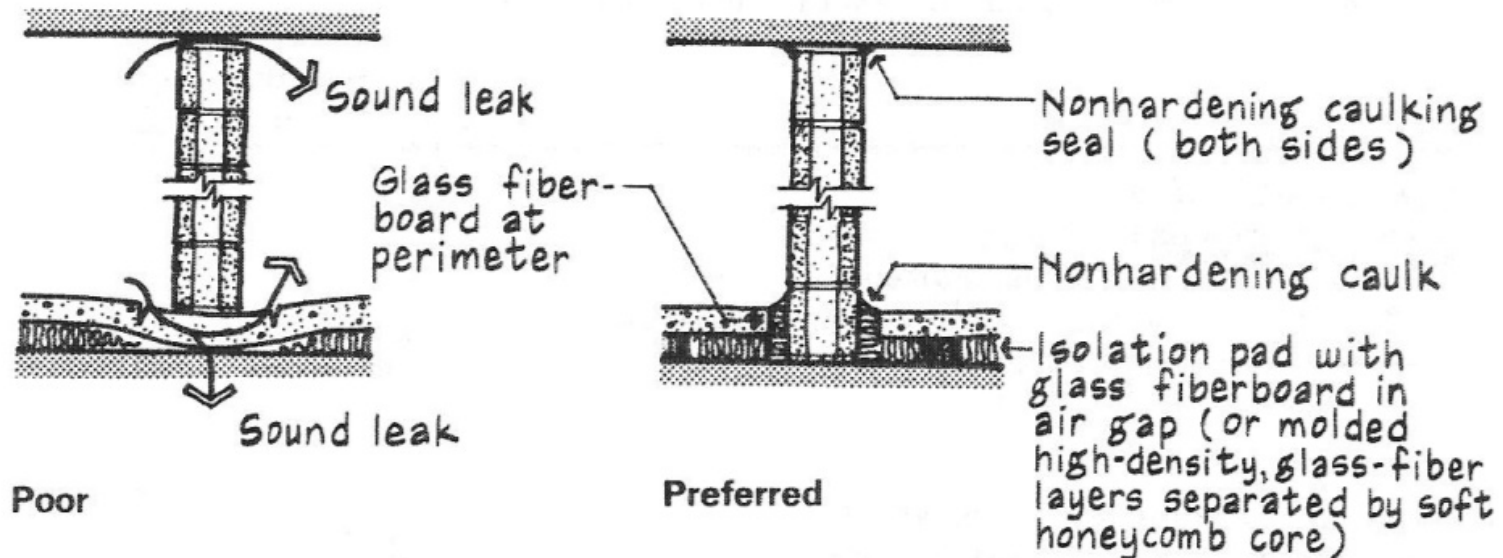


Discontinuity in Floor/Ceiling

- Transmission of structure-borne sound decreases little with distance through materials with little internal damping (e.g., concrete, steel, masonry...structural building elements!)
- Consequently, structure-borne sound can radiate as airborne sound far from the source, particularly if lightweight building materials are encountered
 - Example: A transformer is located on the 28th floor of a building. The characteristic buzz can be heard three floors below.
 - Example: A non-isolated pump was causing problems three buildings away when the vibration reached the condominium gypsum board walls.
 - Example: High fidelity recordings in a concert hall ran into problems when a new air handling unit was installed in a building across the street.

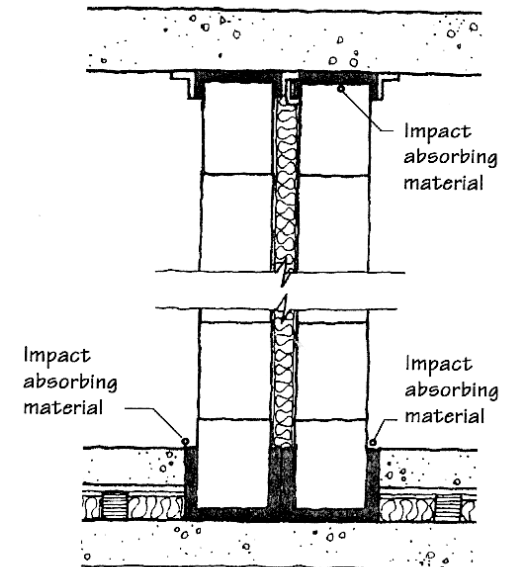
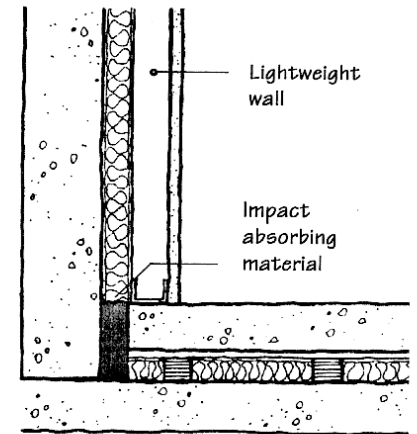
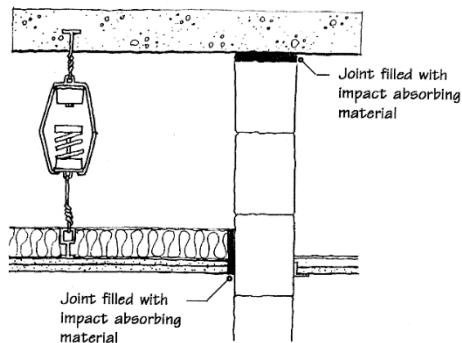
Discontinuity in Floor/Ceiling

- A discontinuity or break provides attenuation of structure-borne sound along its path
 - Physical break, such as a building isolation joint
 - Impact absorbing material, such as resilient isolation mounts



Discontinuity in Floor/Ceiling

- Ceilings: perimeter of resiliently mounted ceiling must have resilient material in joint between ceiling and wall (neoprene most common)
- Floors: perimeter isolation board used to separate floor from wall
- Floors: resilient mounts under floor
- Walls: can be isolated at header and footer
- Penetrations: pipes, ducts, etc. must not touch resiliently mounted construction, or be resiliently mounted themselves



Rain Noise on Roofs

- Often overlooked
- Not much test data available
- Lightweight structures more susceptible
 - Metal roof
 - Membrane roof
- Control strategies
 - Use heavy materials (concrete, roof ballast)
 - Use decoupling material (compliant)
 - Use damped materials (skylights: laminated glass instead of float glass)

Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4